

Coding & Solution

Date	1 July 2026
Team ID	SWTID-2026-3658
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	github.com/Gayathritata/EduGenie.git
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	FastAPI, Jinja2, SQLAlchemy
Key Features Implemented	Secure User Registration & Login (JWT + bcrypt), AI Question Answering, Concept Explanation (LaMini-Flan-T5), Quiz Generation, Text Summarization, Personalized Learning Path Recommendation, Learning History, Bookmarks, SQLite Database Integration, Swagger API Documentation.
Pending / Incomplete Features	Cloud Deployment, Docker Support, PDF Upload (RAG), Voice Assistant, Multilingual Support, Analytics Dashboard
Setup / Run Instructions	1. Clone repository. 2. Create virtual environment. 3. Install dependencies using <code>pip install -r requirements.txt</code>. 4. Configure the <code>.env</code> file with API keys. 5. Run backend using <code>python -m uvicorn app.main:app --reload</code>. 6. Open <code>http://127.0.0.1:8000</code> to access the application.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The EduGenie project follows a clean modular architecture with separate layers for APIs, services, database models, authentication, frontend templates, and static resources. Google Gemini 1.5 Flash and LaMini-Flan-T5 are integrated through reusable service modules, while SQLite and SQLAlchemy manage persistent data storage. The application has been tested successfully, all core features are functioning correctly, and the project is ready for demonstration and future enhancements such as cloud deployment, Retrieval-Augmented Generation (RAG), and multilingual support.